Introduction to Detect-and-Avoid (DAA)

CEC 300: Unmanned Aircraft Design

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Lecture Outline

- Learning Outcomes
- Overview of DAA and key terminology
- Regulatory basis for DAA
  - Part 91
  - Part 107
- Elements of DAA
- Challenges of DAA
- Path Forward
- Review

Learning Outcomes

- Define the role of detect-and-avoid for a UAS system.
- Explain the regulatory need for a detect-and-avoid system to maintain desired level of safety including UAS.
- Identify the major elements of a DAA system
- Identify major challenges with DAA system requirements, design, and capability.

What is Detect and Avoid?

Detect and Avoid

- UAS must be capable of detect and avoiding a potential collision

- Layered approach:
  - **Conflict Avoidance**
    - **Procedural** – procedures in place to avoid collision scenarios within ATC
    - **Air Traffic Management** – provide instruction such as heading or altitude change to avoid a future collision
    - Pilot or Automation – remain well clear
  - **Collision Avoidance**
    - Traditional aircraft – pilot using sight, collision avoidance system (e.g. TCAS II, ACAS X, etc.)
    - Uncrewed – automated or remotely piloted avoidance
      - ACAS Xu

- UAS Detect-and-Avoid addresses the **cooperative and non-cooperative** collision avoidance through
  - **Remain Well Clear** - changes to course to avoid a collision avoidance maneuver
  - **Collision avoidance** – a maneuver to mitigate the risk of a mid-air collision

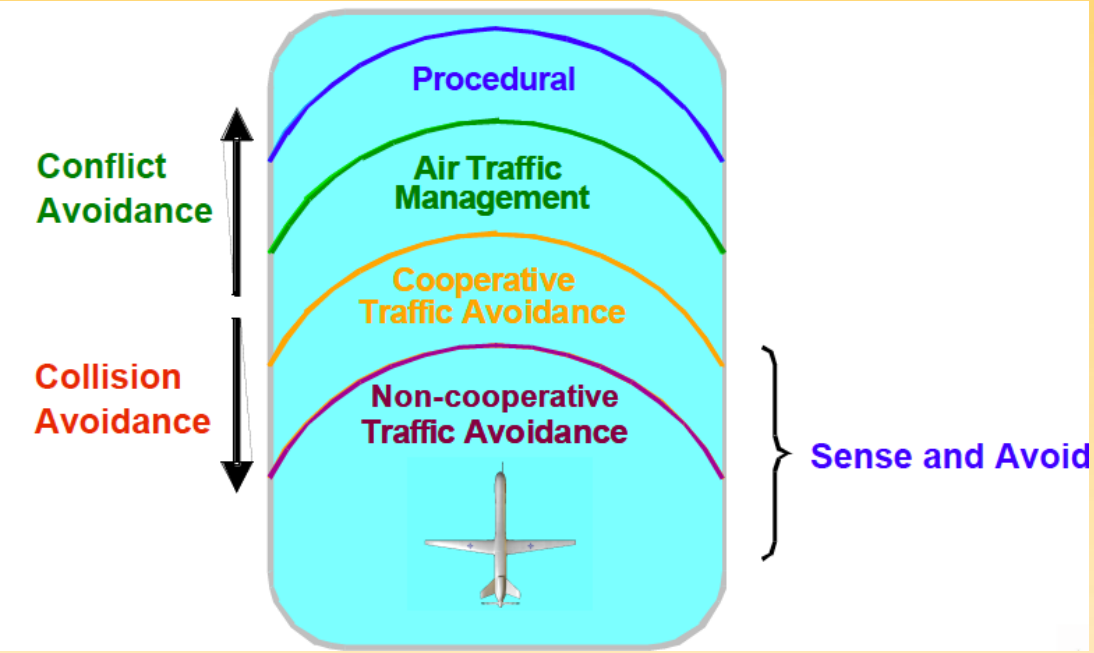


Image Source: NASA Access 5, Collision Avoidance: Functional Requirements (2006)

Types of Collisions

- Mid-Air Collision (MAC) – **PRIMARY FOCUS**
  - Most collision avoidance systems seek to avoid a condition called a Near Mid-Air Collision (NMAC)
- Terrain Collision (see TAWS)
  - Collision between the aircraft and ground
- Runway Incursion
  - Collision between aircraft on runway
- Collision with Objects on Ground

Detect and Avoid is a safety critical function of the system.

Degree of criticality is dependent upon several factors.

Notes on Terminology

- For manned aviation, the term "**see and avoid**" is used within the regulations
  - Qualitative
- For UAS, the "equivalent" capability's terminology has evolved over time:
  - "see and avoid (SAA)"
  - "detect, sense, and avoid (DSA)"
  - "sense and avoid (SAA)"
  - "sense & avoid (S&A)"
  - "detect and avoid (DAA)"
- DAA is the official terminology recognized by the FAA (but they sometimes goof) and we will use it for this course.

Notes on terminology (2)

- From RTCA DO-365:
  - **<u>Collision Avoidance:</u>**
    - "Upper case" – "A last-resort method of preventing mid-air collisions between aircraft, as directed by a Collision Avoidance System."
    - "Lower case" – "A general term referring to the act of maneuvering to prevent mid-air collisions, not tied to any specific Collision Avoidance System"
  - **<u>Collision Avoidance Function</u>** – function performed to achieve lower case definition of collision avoidance
  - **<u>Collision Avoidance System</u>** – a system that produces directive guidance in the for of Resolution Advisories to avoid mid-air collision with other aircraft.
    - E.g. TCAS, TCAS II, ACAS X, etc. - more on these later

Notes on terminology (3)

- More from RTCA DO-365
  - **<u>Self-Separation</u>** – "a system function where the UA maneuvers within a sufficient timeframe to ***remain well clear***."
    - Legacy terminology replaced with the more quantitatively based…
  - **<u>Remain Well Clear (RWC)</u>** – "the ability to detect, analyze and maneuver to avoid potential conflicting traffic by applying adjustments to the current flight path in order to prevent the conflict from developing into a collision hazard"
    - Two aspects:
      - Maintain DWC
      - Regain DWC
  - **<u>DAA Well Clear (DWC)</u>** – a temporal and/or spatial boundary around the aircraft intended to be an electronic means of avoiding conflicting traffic.
    - We shall discuss what well clear means later in this module.
    - Systems like TCAS use some definition of tau (τ), an estimate of time to collision.

Notes on terminology (4)

- In DAA terminology, we separate "our" aircraft from other aircraft with the following terminology:
  - **Ownship** – in a DAA system, ownship refers to the aircraft that will be performing in the detect and avoid functions relative to other aircraft.
  - **Intruder** – in a DAA system, intruder refers to the other aircraft within close enough proximity such that it is tracked and potentially avoiding (RWC or CA).

Example: TCAS II

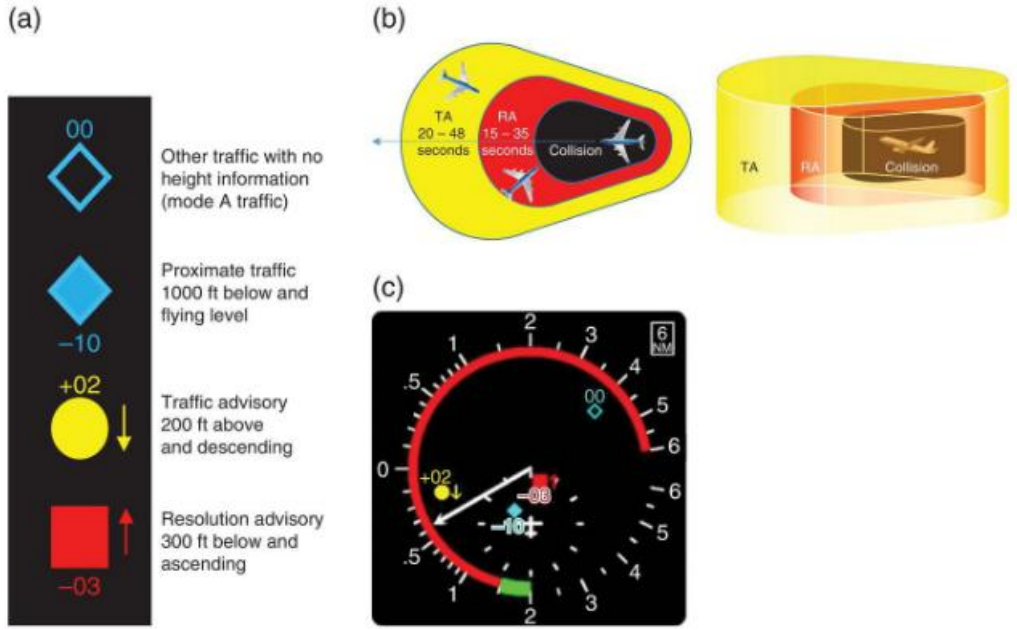


Figure 5.30 (a) Symbols used to depict the four levels of threat in a TCAS display along with the relative height in hundreds of feet and climb or descent. (b) The time to closest point of approach (CPA) and a schematic representation of the volume of space involved to issue a traffic advisory (TA) or resolution advisory (RA). (c) Target vertical speeds required to avoid a collision are indicated by the green range on the display.

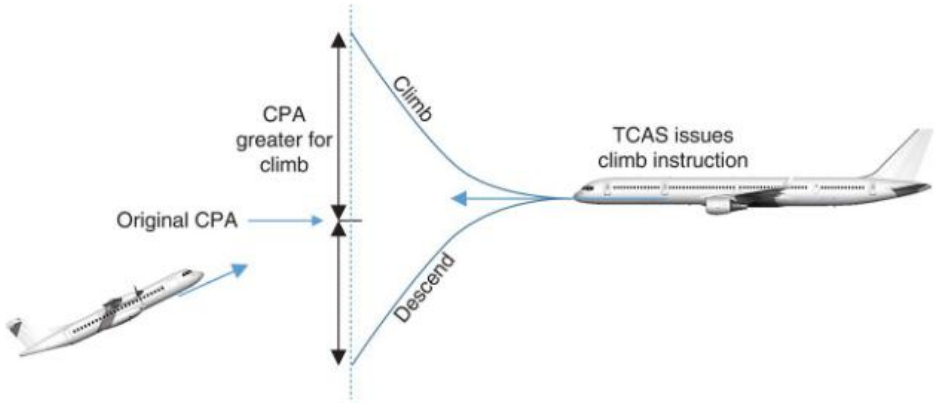


Figure 5.31 The sense of the RA is decided by the option that gives the greatest CPA.

Images from Binns
@ https://learning.oreilly.com/library/view/aircraft-systems/9781119259541/c05.xhtm

Sensor	Application	Functions	Output	Action
TCAS I	General Aviation Use	•Sends signal and interrogates mode C Transponders	•Approximate bearing and relative altitude of all a/c within 40 miles •Color coded dots indicating a/c posing threat – Traffic Advisory (TA)	•Pilot deviate + or – 300 feet ; no lateral •Instrument conditions – notify ATC for assistance
TCAS II	All US commercial carriers	•TCAS I+ •Target intruder tracked and ident a/c is climbing, diving or in straight and level flight	•TCASI + •Target intruder tracked and ident a/c is climbing, diving or in straight and level flight •Resolution Advisory (RA)	•Advise on evasive maneuver for deconfliction •Resolution Advisory (RA) -Preventative – no action -Positive – climb descend 2500 climb or descend

From: NIAG Report SG-205

TCAS will be discussed further in Part II.

Airspace

Airspace	ATC Services
A	• IFR only unless authorized
B	• IFR and VFR (with clearance) • VFR sequencing and separation (limits based upon size) – pilot must still see and avoid • Must have equipage of two-way radio and transponder (unless otherwise authorized)
C	• IFR and VFR (with clearance) • Must have equipage of two-way radio and Mode C or higher transponder (unless otherwise authorized) • VFR sequencing and ATC radar services (radar services outside of boundary if workload permits)
D	• Must have equipage of a two-way radio • No separation services provided to VFR
E	• No specified equipment
G	• Uncontrolled • Supports IFR and VFR operation

New entrants include UAS and UAM

What about segregated airspace?

- The broad consensus from FAA and other CAAs is to avoid implementing UAS or UAM segregated airspace
  - Avoid showing favoritism to one user class
  - Potentially harmful impact to commercial operations

- NASA's UAS Traffic Management (UTM) program has explored low-altitude airspace that may be segregated for UAS operation
  - Looking at < 400 ft. (primarily)
  - e.g. UAS lanes of travel (vertical and horizontal)
  - Still in early technology readiness level (TRL) – concept phase
  - Must identify how to integrate with other low altitude users:
    - Helicopter
    - Ultralight
    - Model aircraft
    - Etc.

- Advanced Air Mobility (AAM) and Urban Air Mobility (UAM) have similar challenges and receive support through Providers of Service for UAM (PSUs) building upon the UTM concepts developed for UAS.
  - No segregation, but coordination for non-UAM to enter UAM corridor and vice versa.

Need for UAS DAA

Regulatory Basis from Manned

- Title 14 Code of Federal Regulations Part 91, sections:
  - 91.111 Operating near other aircraft
  - 91.113 Right-of-way rules: except water operations
  - 91.115 Right of way rules: water operations
  - 91.181 Course to be flown

- For uncrewed systems (UAS and UAM), the function must be reallocated from an onboard pilot to human-in-the-loop or human-on-the-loop automation, and autonomous mode under lost link conditions.
- An equivalent level of safety is required to be certifiable.

Section 91.111

§91.111 Operating near other aircraft.

(a) No person may operate an aircraft so close to another aircraft as to create a collision hazard.

(b) No person may operate an aircraft in formation flight except by arrangement with the pilot in command of each aircraft in the formation.

(c) No person may operate an aircraft, carrying passengers for hire, in formation flight.

- Forbids operator of controlling their aircraft in close proximity to other aircraft.
- Maneuver cannot be made if a collision hazard exists.
- Impact:
  - Manned aviation requires self-separation by pilot to mitigate risk of collision
  - Unmanned aviation requires some self-separation mechanism built into the UAS to mitigate risk of collision
    - Sufficient accuracy and resolution
    - Operator or automation must be able to determine:
      - Location of ownship
      - Location and trajectory of intruder
      - Pilot/automation must maneuver aircraft to maintain safe separation
  - Qualitative and quantitative requirements and data for DAA avionic components to ensure sufficient level of safety

Section 91.113

§91.113 Right-of-way rules: Except water operations.

(a) *Inapplicability.* This section does not apply to the operation of an aircraft on water.

(b) *General.* When weather conditions permit, regardless of whether an operation is conducted under instrument flight rules or visual flight rules, vigilance shall be maintained by each person operating an aircraft so as to see and avoid other aircraft. When a rule of this section gives another aircraft the right-of-way, the pilot shall give way to that aircraft and may not pass over, under, or ahead of it unless well clear.

(c) *In distress.* An aircraft in distress has the right-of-way over all other air traffic.

(d) *Converging.* When aircraft of the same category are converging at approximately the same altitude (except head-on, or nearly so), the aircraft to the other's right has the right-of-way. If the aircraft are of different categories—

(1) A balloon has the right-of-way over any other category of aircraft;

(2) A glider has the right-of-way over an airship, powered parachute, weight-shift-control aircraft, airplane, or rotorcraft.

(3) An airship has the right-of-way over a powered parachute, weight-shift-control aircraft, airplane, or rotorcraft.

However, an aircraft towing or refueling other aircraft has the right-of-way over all other engine-driven aircraft.

(e) *Approaching head-on.* When aircraft are approaching each other head-on, or nearly so, each pilot of each aircraft shall alter course to the right.

(f) *Overtaking.* Each aircraft that is being overtaken has the right-of-way and each pilot of an overtaking aircraft shall alter course to the right to pass well clear.

(g) *Landing.* Aircraft, while on final approach to land or while landing, have the right-of-way over other aircraft in flight or operating on the surface, except that they shall not take advantage of this rule to force an aircraft off the runway surface which has already landed and is attempting to make way for an aircraft on final approach. When two or more aircraft are approaching an airport for the purpose of landing, the aircraft at the lower altitude has the right-of-way, but it shall not take advantage of this rule to cut in front of another which is on final approach to land or to overtake that aircraft.

- Addresses **necessary vigilance** required by the system (operator and/or automation) to see and avoid other aircraft.
- Identifies **right-of-way requirements** concerning a variety of airborne platforms
- Technology is needed to determine two aircraft's **flight paths are converging unsafely**
- Current gaps for automated operations:
  - Landing requirements state that lower aircraft has right of way
    - Must be able to determine intent to land from other aircraft
    - Must have a sensing field of view that includes a sufficient vertical component to detect aircraft above and below
  - Distress situations give right-of-way to aircraft in distress
    - Automation or PIC must have sufficient situational awareness (e.g. voice communications) to be informed of the distress situation.
    - Sensing must be capable of allowing automation or PIC to identify the distressed aircraft and yield

Section 91.115

§91.115 Right-of-way rules: Water operations.

(a) *General.* Each person operating an aircraft on the water shall, insofar as possible, keep clear of all vessels and avoid impeding their navigation, and shall give way to any vessel or other aircraft that is given the right-of-way by any rule of this section.

(b) *Crossing.* When aircraft, or an aircraft and a vessel, are on crossing courses, the aircraft or vessel to the other's right has the right-of-way.

(c) *Approaching head-on.* When aircraft, or an aircraft and a vessel, are approaching head-on, or nearly so, each shall alter its course to the right to keep well clear.

(d) *Overtaking.* Each aircraft or vessel that is being overtaken has the right-of-way, and the one overtaking shall alter course to keep well clear.

(e) *Special circumstances.* When aircraft, or an aircraft and a vessel, approach so as to involve risk of collision, each aircraft or vessel shall proceed with careful regard to existing circumstances, including the limitations of the respective craft.

- Similar to 91.113, but also addresses right-of-way for water operations including non-aircraft vessels
  - Head-on, crossing, and overtaking conditions
- Must be addressed for UAS designed for water takeoff/landing
- Visual observation can be more difficult with water operations
- Identifying surface vessel capabilities and intent is also required by sensors, visual observers, or PIC.

Section 91.181

§91.181 Course to be flown.

Unless otherwise authorized by ATC, no person may operate an aircraft within controlled airspace under IFR except as follows:

(a) On an ATS route, along the centerline of that airway.

(b) On any other route, along the direct course between the navigational aids or fixes defining that route. However, this section does not prohibit maneuvering the aircraft to pass well clear of other air traffic or the maneuvering of the aircraft in VFR conditions to clear the intended flight path both before and during climb or descent.

- Addresses permission of a pilot to maneuver to pass ”well clear” of other air traffic within controlled airspace.

Other FAA Documents

- Advisory Circulars
  - AC 90-48c – Pilot's role in collision avoidance
  - AC 91-73c – Single pilot taxi operations
- FAA Orders
  - FAA Order 7610.4k – Military operations, includes requirements for equivalent level of safety in see and avoid
  - FAA Order 8130.34 – UAS airworthiness, includes requirements for visual observation
- Other:
  - FAA Aeronautical Information Manual (AIM)
  - FAA-H-8083-3A: Airplane Flying Handbook (see and avoid, runway incursion avoidance)

DAA Requirement

§107.37 Operation near aircraft; right-of-way rules.

(a) Each small unmanned aircraft must yield the right of way to all aircraft, airborne vehicles, and launch and reentry vehicles. Yielding the right of way means that the small unmanned aircraft must give way to the aircraft or vehicle and may not pass over, under, or ahead of it unless well clear.

(b) No person may operate a small unmanned aircraft so close to another aircraft as to create a collision hazard.

Visual Line of Sight / Observers

§107.31 Visual line of sight aircraft operation.

(a) With vision that is unaided by any device other than corrective lenses, the remote pilot in command, the visual observer (if one is used), and the person manipulating the flight control of the small unmanned aircraft system must be able to see the unmanned aircraft throughout the entire flight in order to:

(1) Know the unmanned aircraft's location;

(2) Determine the unmanned aircraft's attitude, altitude, and direction of flight;

(3) Observe the airspace for other air traffic or hazards; and

(4) Determine that the unmanned aircraft does not endanger the life or property of another.

(b) Throughout the entire flight of the small unmanned aircraft, the ability described in paragraph (a) of this section must be exercised by either:

(1) The remote pilot in command and the person manipulating the flight controls of the small unmanned aircraft system; or

(2) A visual observer.

§107.33 Visual observer.

If a visual observer is used during the aircraft operation, all of the following requirements must be met:

(a) The remote pilot in command, the person manipulating the flight controls of the small unmanned aircraft system, and the visual observer must maintain effective communication with each other at all times.

(b) The remote pilot in command must ensure that the visual observer is able to see the unmanned aircraft in the manner specified in §107.31.

(c) The remote pilot in command, the person manipulating the flight controls of the small unmanned aircraft system, and the visual observer must coordinate to do the following:

(1) Scan the airspace where the small unmanned aircraft is operating for any potential collision hazard; and

(2) Maintain awareness of the position of the small unmanned aircraft through direct visual observation.

FAA Part 108 (pending)

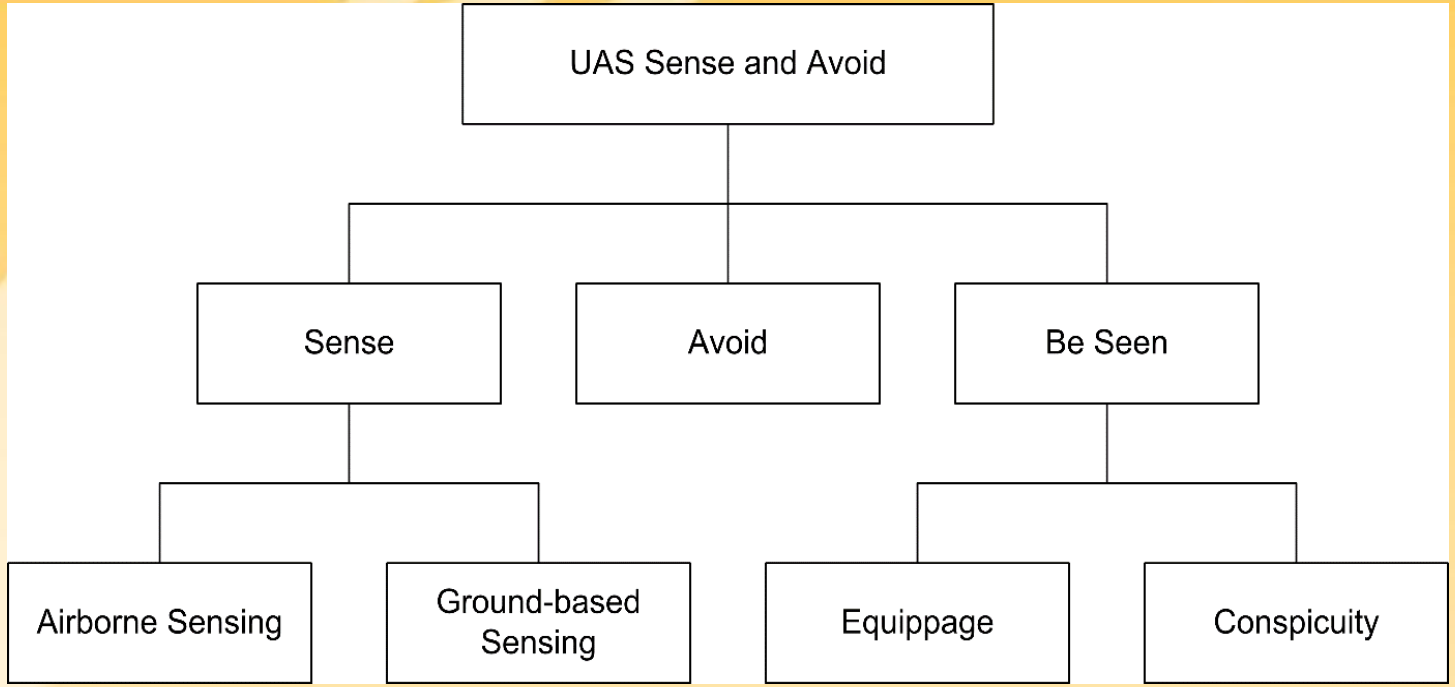
- An aviation rule making committee provided a report with recommendations for Beyond Visual Line of Sight Operations (BVLOS).
  - Expectation of equipage with either ADS-B or TABS (Traffic Awareness Beacon System)
  - Identifies rules and procedures for BVLOS including
    - Shielded operations – operations in which the aircraft is shielded from other aircraft based on the environment, e.g. urban canyons.
    - Non-shielded operations – typical operations

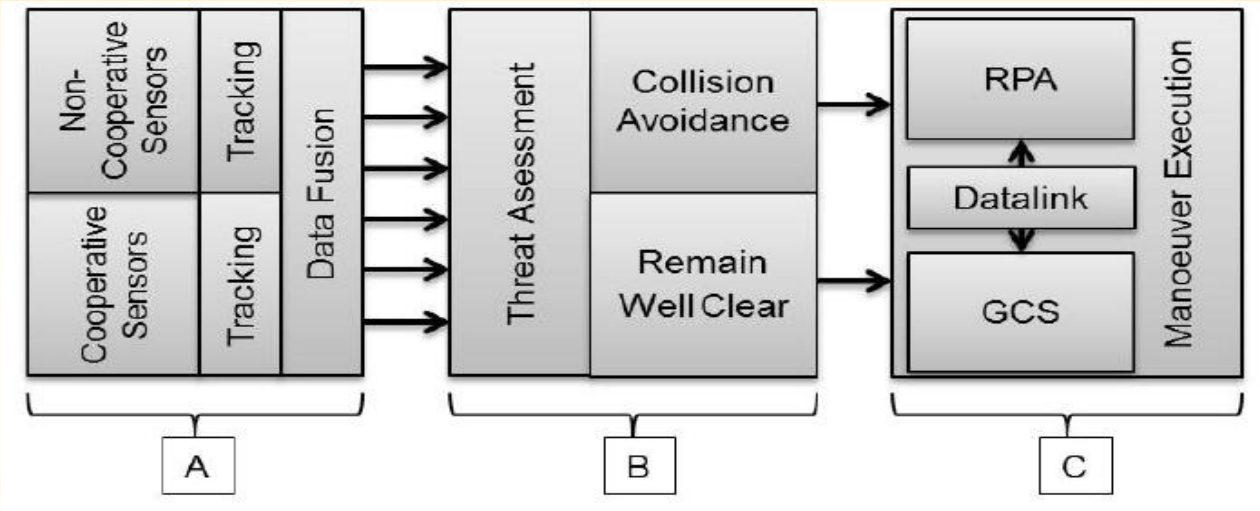
Elements of DAA and Definitions

Pilot's Role

- We cannot assume that the human operator (pilot in command) does not have a role in DAA.
- Notional examples:
  - Human-in-the-loop:
    - Processes sensor inputs
    - Command aircraft to performance avoidance
  - Human-on-the-loop:
    - DAA system detects potential collision threat and notifies pilot
    - Pilot approves recommended course of action and/or inputs a corrective action given the alert
  - Automated:
    - DAA system detects potential collision and takes corrective action
    - PIC is notified of the course change (ability to override?)
    - Necessary to support contingency operations
- Visual observers may be employed instead of sensors as recommended under Part 107
  - Ground-based
  - Airborne
- Pilot in command has final legal authority of the actions of the aircraft
  - In manned aviation, pilot can choose to justifiably disregard a resolution advisory or ATC instruction if they feel it jeopardizes flight safety

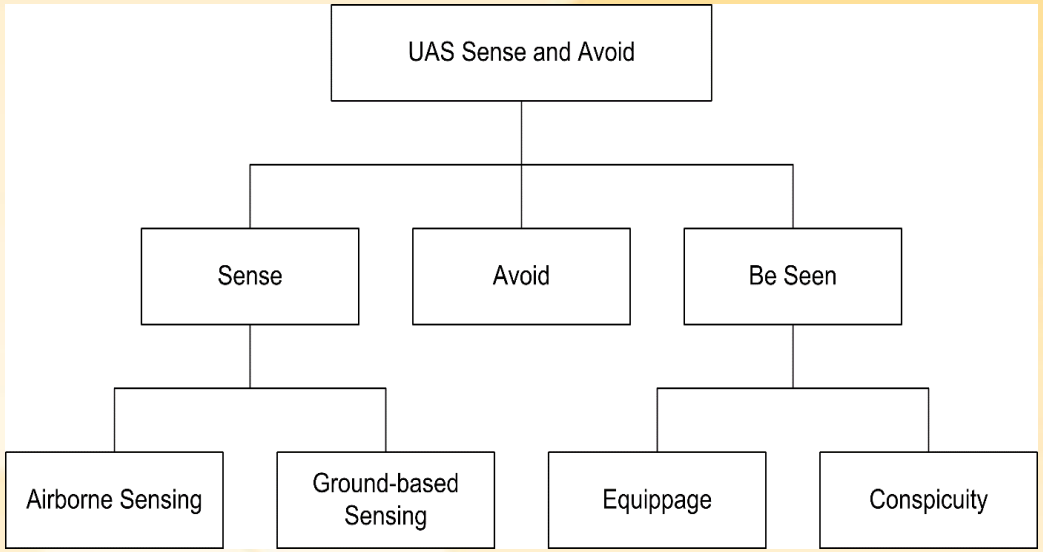
Notional Taxonomy and Generic Architecture





Sense / Detect

- Detect – identification of a collision risk through sensor data
  - Human
  - Machine
  - Human augmenting machine



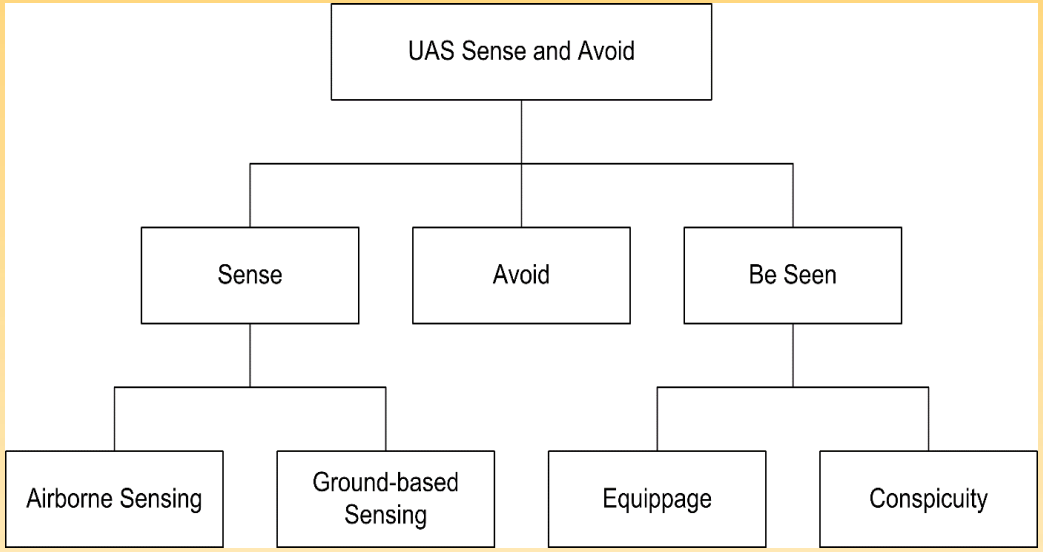
Human Detection (Detect)	The ability of the human operator to harness their five senses in an effort to detect aircraft. This includes visual detection and noise detection.
Machine Detection	The incorporation of machine-based systems that enhance the operator's situational awareness in an effort to prevent collisions.

Sense / Detect (2)

- Cooperative vs. Non-Cooperative Aircraft
  - One major distinction in modalities for detection is whether or not the “intruder” is a cooperative aircraft vs. a non-cooperative aircraft
  - **Cooperative aircraft:**
    - Equipped with a form of cooperative technology such as:
      - Transponder
      - ADS-B
      - TCAS /ACAS I or II
      - Identify, Friend, or Foe System
      - Etc.
      - Note: more on these later
    - Required for operation in controlled airspace / under an instrument flight rules (IFR) flight plan
  - **Non-cooperative aircraft:**
    - Not equipped with a technology that aids in detection through traditional air traffic management system, transponder system, etc.
    - Non-cooperative can utilize other means to help ”be seen”
    - Permissible for some aircraft types and/or some airspace categories
- Non-cooperative aircraft are one of the great challenges of UAS integration

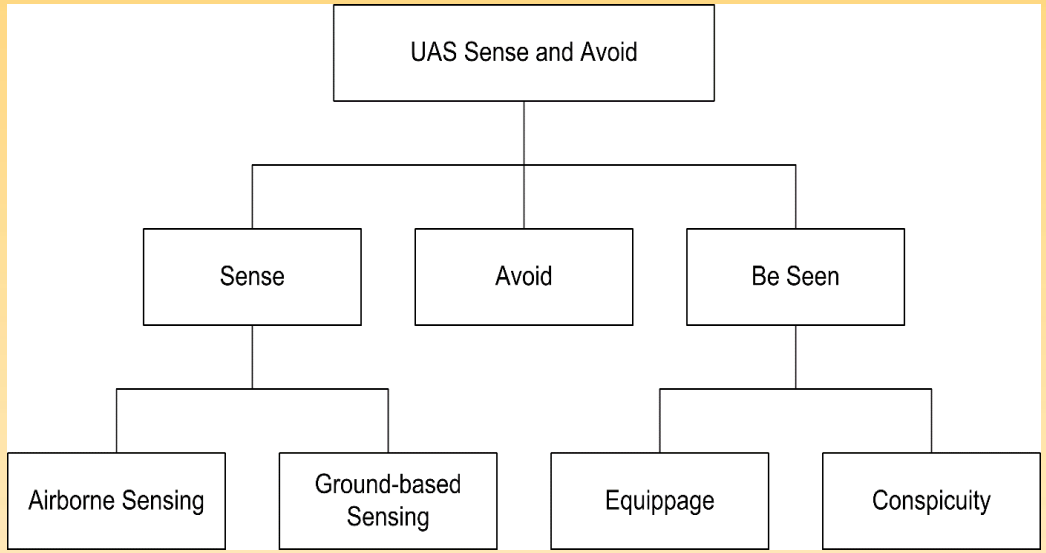
Sense / Detect: Airborne

- Sensors are integrated into the UAS
- Cooperative vs. non-cooperative sensing options
- Advantages:
  - Mature for larger systems
  - Opportunity for greater independence with control station
  - Supports a broader area of operation
- Disadvantages:
  - Size, weight, and power restrictions can limit options for small UAS
  - Small UAS technologies typically have a lower technical readiness level (TRL)



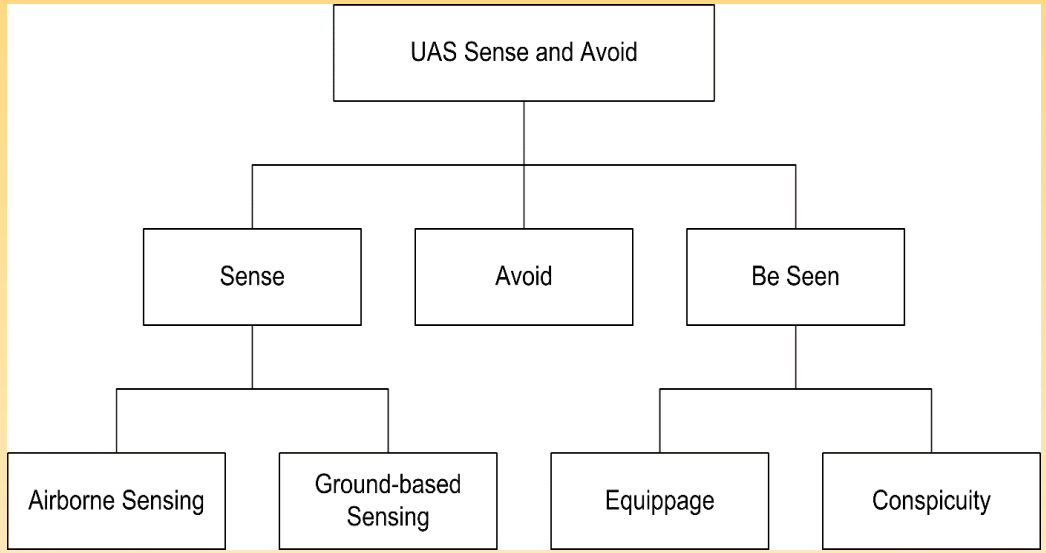
Sense / Detect: Ground-based

- Sensors are located on the ground
  - Fixed
    - UAS specific
    - Existing infrastructure
  - Portable
- Advantages:
  - Reduces/eliminates SWAP constraints
  - Can leverage existing technologies
- Disadvantage:
  - System now dependent upon command-and-control data link



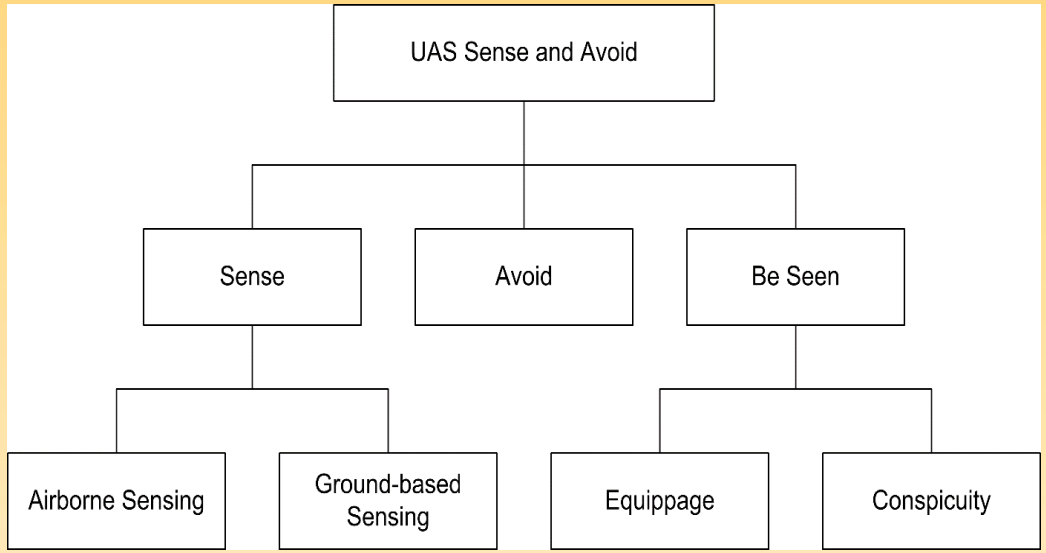
Avoid

- UAS must make corrective action:
  - Remain Well Clear
    - Maintaining a safe separation with other aircraft
    - Can be handled procedurally (e.g. ATC controlled airspace)
  - Collision avoidance
    - Maneuver is made to ensure the UAS remains well clear of other aircraft
- The delineation between these activities varies significantly between proposed approaches
- Standardization of “well clear” is vital element, which will be discussed further



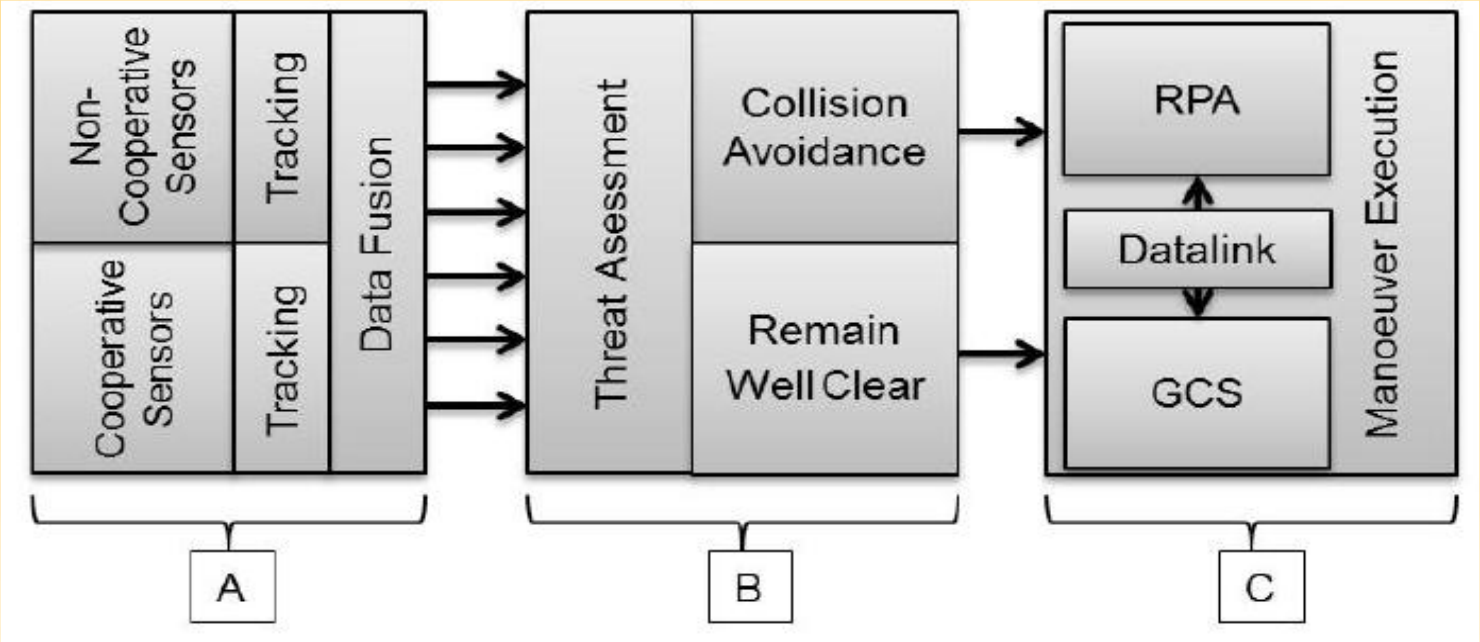
Be Seen

- Current regulatory trends toward UAS detect-and-avoid falling under responsibility of UAS
- However, the ability of the UAS to be seen is of key importance:
  - Be seen by other air traffic
  - Be seen by visual observers
- Approaches:
  - Equipage: make the UAS a cooperative aircraft
  - Conspicuity: enhance the UAS to aid in detection
    - Paint, lights, RF reflecting paint/materials, etc.



Other Elements of Consideration

- Architecture
  - Ground-based, Airborne, or Mixed
  - Pilot-in-the-loop, Pilot-on-the-loop, Automated
- Sensor fusion
- Intruder tracking



Challenges of DAA

Challenges for UAS DAA

- Creating quantifiable requirements for UAS DAA
  - How do we create an “equivalent level of safety” when Part 91 criteria are subjective?
  - Includes:
    - Human-based
    - Machine-based
  - Sensing requirements
  - Definition of “well clear”
- Identify requirements of UAS DAA Architecture(s)
  - Address architectural needs and requirements of UAS DAA
    - Ground-based
    - Airborne
    - Mixed Air/Ground
  - Understanding the limits of sensor types and creating a complimentary suite of sensors

“Swiss Cheese” Analogy

- Detect and avoid sensing has been described as working with “swiss cheese”
- Sensors and tracking software might have “holes” in its safety function.
  - Detection limits can include, but are not limited to: range, minimum target size, minimum target velocity, altitude of ownship, altitude of intruder, etc.
- A goal of DAA sensor selection is to find the right layers of “cheese” such that when stacked together no holes remain.

Road Ahead for DAA Lecture

Path forward for DAA Lectures

EMBRY-RIDDLE Aeronautical University

Upcoming Topics:

- Sensor Technologies
  - Airborne Sensors
  - Ground-based Sensors
- Defining Well Clear
- Avoidance Algorithms
- DAA Architectural Considerations
- Regulatory Considerations

Review

- A DAA system is designed to ensure that a UAS remains well clear of other aircraft, and performs collision avoidance maneuvers if a well clear violation occurs or is eminent.
- DAA requirements must be developed to meet the expectations of 14 CFR Part 91 to see-and-avoid and follow right-of-way rules within air traffic.
- Part 107 for small UAS relies upon visual observers, but waivers permit beyond visual line of sight operations, which require a DAA system
- Elements of the DAA system were summarized with more detail ahead
- Numerous challenges exist with DAA systems including establishing performance metrics, defining system requirements, and ensuring adequate detection capabilities.

References

- FAA Aeronautical Information Manual: https://www.faa.gov/air_traffic/publications/media/aim.pdf
- RTCA DO-365: Minimum Operational Performance Standards (MOPS) for Detect and Avoid (DAA) Systems
- Title 14 CFR Part 91
- Title 14 CFR Part 107
- NATO Industrial Advisory Group (NIAG) Study Group 205, "Final Report on NIAG Study Group 205 on Sense and Avoid Feasibility and Certification for UAS Flight in Non-Segregated Airspace"
- Stansbury and Wilson, "Technology Surveys and Regulatory Gap Analyses of UAS Sub-Systems toward Access to the NAS," In the *Handbook for Unmanned Aerial Vehicles*, 2014.